

px and then processed by seven model blocks that are extracting multi-scale contextual information and generating a high-resolution disparity map. Each block consists of three convolution layers with one additional skip connection. At the end of the decoder module, the obtained disparity map is superimposed with the original image to get the final bokeh output. The model was trained with a combination of the L1, L2 and SSIM loss functions. Network parameters were optimized using the Adam algorithm with a learning rate of $5e-5$ and a batch size of 8.

5 Additional Literature

An overview of the past challenges on mobile-related tasks together with the proposed solutions can be found in the following papers:

- Learned End-to-End ISP: [29,33,23,19]
- Perceptual Image Enhancement: [32,24]
- Bokeh Effect Rendering: [22,31]
- Image Super-Resolution: [32,47,4,60]

Acknowledgements

We thank the sponsors of the Mobile AI and AIM 2022 workshops and challenges: AI Witchlabs, MediaTek, Huawei, Reality Labs, OPPO, Synaptics, Raspberry Pi, ETH Zürich (Computer Vision Lab) and University of Würzburg (Computer Vision Lab).

A Teams and Affiliations

Mobile AI 2022 Team

Title:

Mobile AI 2022 Learned Smartphone ISP Challenge

Members:

Andrey Ignatov^{1,2} (andrey@vision.ee.ethz.ch), Radu Timofte^{1,2,3}

Affiliations:

¹ Computer Vision Lab, [ETH Zurich](#) Switzerland

² AI Witchlabs, Switzerland

³ [University of Wuerzburg](#) Germany

Antins_cv***Title:***

A Tiny UNet for Image Bokeh Rendering

Members:

Jin Zhang (*zj346862@antgroup.com*), Feng Zhang, Gaocheng Yu, Zhe Ma, Hongbin Wang

Affiliations:

Ant Group, China

ENERZAI***Title:***

Bokeh Unet and Histogram Feature Loss

Members:

Minsu Kwon (*minsu.kwon@enerzai.com*)

Affiliations:

ENERZAI, South Korea

ZJUT-Vision***Title:***

Attention-based Double V-Net

Members:

Haotian Qian (*1092944263@qq.com*), Wentao Tong, Pan Mu, Ziping Wang, Guangjing Yan

Affiliations:

Zhejiang University of Technology, China

Sensebrain***Title:***

Bokeh-Loss GAN: Multi-stage Adversarial Training for Realistic Edge-Aware Bokeh [42]

Members:

Brian Lee¹ (*brianlee@sensebrain.site*), Lei Fei², Huaijin Chen¹

Affiliations:

¹ Sensebrain Technology, United States

² Tetras.AI, China